

Primary Aldosteronism — AVS, Subtyping & Treatment

DETECTIVE BULLETIN BOARD

ARR SCREENING LAB

CONFIRMATION AND IMAGING FLOOR

TREATMENT ARMORY

WHEN TO SCREEN

ARR SCREENING LAB

CONFIRMATION TEST BENCH

OPERATING THEATRE

MEDICATION INTERFERENCE WALL

TABLE 398-4 ARR INTERFERENCE

TABLE 398-5 ADRENAL MASS CLASSIFICATION

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CLINICAL SUSPICION BULLETIN BOARD

PREPARATION STATION

UNENHANCED ADRENAL CT SCANNER

MEDICAL THERAPY LADDER

CLINICAL PEARL: SCREEN HIGH RISK WITH ARR. CONFIRM AUTONOMY. CT THEN AVS IF SURGICAL CANDIDATE OR AGE OVER 35. UNILATERAL EQUALS ADRENALECTOMY. BILATERAL EQUALS SPIRO OR EPLERENONE.

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1. When to screen

WHAT YOU SEE

CLINICAL SUSPICION BULLETIN: resistant HTN, HTN + hypokalemia, adrenal incidentaloma, early stroke/onset <40, family history

WHAT IT ENCODES

Screen for primary aldosteronism in: resistant HTN (uncontrolled on ≥ 3 drugs incl. a diuretic), HTN with spontaneous or diuretic-induced hypokalemia, HTN + adrenal incidentaloma, HTN with onset <40 or early stroke, OSA, and HTN with a family history of early PA or stroke. PA is the most common SECONDARY/curable cause of HTN (~5–10% of hypertensives).

BOARD TRAP

Restricting screening to hypokalemic patients — the majority of PA patients are NORMOKALEMIC, so requiring low K⁺ misses most cases.

2. Prep: correct K, stop spiro

WHAT YOU SEE

PREPARATION STATION: potassium back up, stop spiro 4-6 weeks, STOP sign

WHAT IT ENCODES

Before measuring the ARR: correct hypokalemia (low K⁺ falsely suppresses aldosterone and lowers the ratio) and stop mineralocorticoid-receptor antagonists (spironolactone, eplerenone) and amiloride for 4–6 weeks. Liberalize sodium intake and draw the sample mid-morning after the patient has been up ~2 h.

BOARD TRAP

Drawing the ARR while the patient is still on an MRA or hypokalemic gives a false-negative — never screen without prepping first.

3. ARR screening

WHAT YOU SEE

ALDOSTERONE:RENIN RATIO scale, ARR greater than 750, aldo high / renin suppressed

WHAT IT ENCODES

The screening test is the plasma aldosterone-to-renin ratio (ARR). A positive screen = HIGH aldosterone (typically PAC >15 ng/dL) with SUPPRESSED renin, giving an elevated ratio. The suppressed renin is what distinguishes PRIMARY (autonomous) aldosteronism from secondary hyperaldosteronism (where renin is high).

BOARD TRAP

A high aldosterone alone is not diagnostic — it must be paired with suppressed renin. Secondary hyperaldosteronism (renal artery stenosis, diuretics, CHF) has HIGH renin and a normal ratio.

4. Medication interference wall

WHAT YOU SEE

Shelves: false negatives (MR antagonists, diuretics, ACEi/ARB, SSRI) vs false positives (beta-blockers, clonidine, methyldopa, NSAIDs, estrogen)

WHAT IT ENCODES

Many antihypertensives distort the ARR. FALSE NEGATIVES (lower ARR): MRAs, diuretics, ACE inhibitors/ARBs, dihydropyridine CCBs, SSRIs. FALSE POSITIVES (raise ARR): beta-blockers, central agonists (clonidine, methyldopa), NSAIDs, and estrogens. The safe drugs to continue during testing are non-dihydropyridine effects aside, verapamil-SR, hydralazine, and alpha-1 blockers (e.g. doxazosin/prazosin).

BOARD TRAP

Beta-blockers cause FALSE-POSITIVE ARR (suppress renin more than aldosterone); ACEi/ARBs cause FALSE-NEGATIVES (raise renin). Don't misread an ARR taken on these drugs as definitive — switch to verapamil/alpha-blockers.

5. Confirmatory testing

WHAT YOU SEE

CONFIRMATION TEST BENCH: saline infusion, oral sodium load, fludrocortisone — aldo NOT suppressed

WHAT IT ENCODES

A positive ARR must be CONFIRMED (except in the overt case of spontaneous hypokalemia + suppressed renin + aldosterone >20, which needs no confirmation). Confirmatory tests demonstrate aldosterone AUTONOMY — failure to suppress aldosterone with a saline infusion test, oral sodium loading, fludrocortisone suppression test, or captopril challenge.

BOARD TRAP

The ARR is only a SCREEN; skipping confirmation and going straight to imaging/surgery overcalls PA. Sodium loading is contraindicated in severe/uncontrolled HTN or heart failure.

6. Unenhanced adrenal CT

WHAT YOU SEE

CT scanner: unilateral mass, bilateral hyperplasia, micronodularity

WHAT IT ENCODES

After biochemical confirmation, do a thin-cut adrenal CT to characterize morphology (unilateral adenoma, bilateral hyperplasia, micronodularity) and — importantly — to EXCLUDE adrenocortical carcinoma. CT defines anatomy but does not establish which gland is the functional source.

BOARD TRAP

Relying on CT alone to decide surgery — a CT 'adenoma' may be a non-functioning incidentaloma while the other (CT-normal) gland is the actual source. CT and AVS disagree in ~30–40% of cases.

7. AVS to lateralize

WHAT YOU SEE

OPERATING THEATRE note: bilateral or nonsurgical vs unilateral; AVS needed age ≥ 35

WHAT IT ENCODES

Adrenal vein sampling (AVS) is the GOLD STANDARD to lateralize before surgery — it directly measures aldosterone (cortisol-corrected) from each adrenal vein to prove a unilateral source. Guidelines recommend AVS in all surgical candidates, with the exception that patients <35 with spontaneous hypokalemia, marked aldosterone excess, and a clear unilateral CT adenoma may proceed to surgery without it.

BOARD TRAP

Skipping AVS and operating off CT alone → operating on the wrong adrenal. Conversely, AVS is unnecessary if the patient is not a surgical candidate or doesn't want surgery.

8. Unilateral → adrenalectomy

WHAT YOU SEE

LAPAROSCOPIC ADRENALECTOMY, potentially curative, BP normalizes afterwards

WHAT IT ENCODES

A lateralized UNILATERAL source (Conn adenoma or unilateral hyperplasia) → laparoscopic adrenalectomy, which is potentially curative: hypokalemia resolves in ~100%, and BP improves or normalizes in most (cure of HTN in ~30–60%, better with younger age and shorter HTN duration).

BOARD TRAP

Pre-op, control BP/K+ with an MRA; post-op watch for transient HYPOaldosteronism/hyperkalemia as the chronically suppressed contralateral gland recovers.

9. Medical therapy ladder

WHAT YOU SEE

MEDICAL THERAPY LADDER: spironolactone (gynecomastia/menstrual), eplerenone selective, amiloride for Liddle/hyperkalemia

WHAT IT ENCODES

BILATERAL disease or non-surgical patients → lifelong mineralocorticoid-receptor antagonist. SPIRONOLACTONE is first-line (cheap, effective) but is non-selective → gynecomastia, erectile dysfunction, menstrual irregularity. EPLERENONE is the selective second-line when those side effects occur (fewer anti-androgen effects, but weaker and twice-daily). Amiloride (ENaC blocker) is an add-on/alternative.

BOARD TRAP

Using amiloride as the PRIMARY agent for true primary aldosteronism — first-line is an MRA (spironolactone/eplerenone), not amiloride. Amiloride is the drug of choice for Liddle syndrome, not PA.

10. Adrenal mass: benign

WHAT YOU SEE

TABLE 398-5 ADRENAL MASS CLASSIFICATION, BENIGN: adenoma, myelolipoma, cyst

WHAT IT ENCODES

Benign adrenal masses — adenoma, myelolipoma (fat-containing), simple cyst — are typically small, homogeneous, and LOW attenuation (≤10 HU on unenhanced CT) with rapid contrast washout. These features reassure against malignancy.

BOARD TRAP

A lipid-rich adenoma is ≤10 HU; higher attenuation, large size (>4 cm), or irregular margins should raise concern for ACC or pheochromocytoma.

11. Indeterminate / malignant

WHAT YOU SEE

INDETERMINATE and MALIGNANT rows: pheochromocytoma, metastasis, ACC

WHAT IT ENCODES

Indeterminate/malignant masses — pheochromocytoma, metastasis, adrenocortical carcinoma (ACC) — show high unenhanced attenuation (>20 HU), large size, heterogeneity, and slow washout, requiring further hormonal and imaging workup ± resection.

BOARD TRAP

Always exclude pheochromocytoma biochemically (plasma/urine metanephrines) BEFORE any adrenal biopsy or surgery — biopsying a pheo can precipitate a hypertensive crisis.

12. Hormonal workup of mass

WHAT YOU SEE

GC-MS and 24-hour urinary steroid profile panels

WHAT IT ENCODES

Every adrenal incidentaloma needs a functional workup: 1-mg overnight dexamethasone suppression (autonomous cortisol), plasma/urine metanephrines (pheo), and ARR if hypertensive (aldosterone). Urinary steroid metabolomics by GC-MS/24-h profiling can flag ACC.

BOARD TRAP

Don't forget to screen even an 'asymptomatic' incidentaloma for subclinical cortisol excess and pheo — functional but clinically silent tumors are common.

13. Clinical pearl

WHAT YOU SEE

Bottom banner

WHAT IT ENCODES

One-line algorithm: screen high-risk HTN with ARR (high aldo + low renin) → confirm autonomy (saline/sodium load) → adrenal CT → AVS if surgical candidate (skip only if <35 with classic unilateral picture). Unilateral = laparoscopic adrenalectomy; bilateral/non-surgical = spironolactone (or eplerenone if side effects).

BOARD TRAP

The two highest-yield traps: K+ is usually NORMAL (don't gate screening on hypokalemia), and CT cannot replace AVS for lateralization.